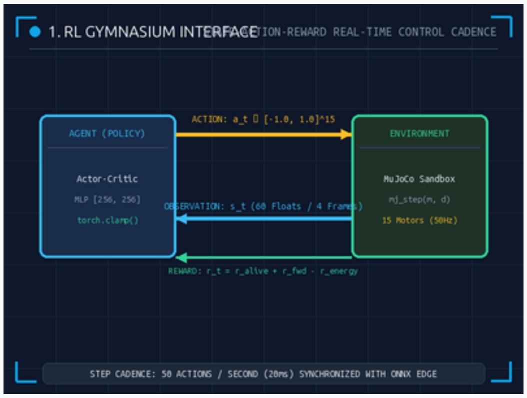
■ MICRODUCK PHYSICAL AI MASTERCLASS

PHASE 3 HANDOUT

Module 3: The Dog Trainer: Gym Environments, Reward Engineering, PPO & Curriculum Learning

RL & PPO

ALGORITHM: PPO (Clipped)	OBSERVATION: 60 Floats (4 Frames)	ACTION: 15 Clamped [-1, 1]	FREQUENCY: 50 Hz Control Loop



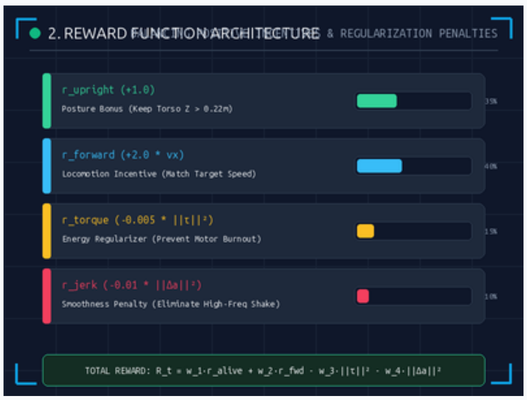
■ 1. Gymnasium Environment & State Loop

OBSERVATION & ACTION CADENCE

The RL Paradigm: The policy observes a sliding history tensor $s_t \in \mathbb{R}^{60}$, infers 15 normalized motor velocity actions $a_t \in [-1.0, 1.0]$, and receives a scalar reward.

The 12-Year-Old Analogy: When you train a puppy to fetch, you don't pick up its paws. You show it the ball, watch what it does, and toss a biscuit when it brings it back.

- **Observation Space**: Box(-inf, inf, shape=(60,), float32) representing 4 frames × 15 sensors.
- **Action Space**: Box(-1.0, 1.0, shape=(15,), float32) target joint velocities.
- **50Hz Execution**: Strict 20ms step cadence ensuring zero sim-to-real timing discrepancy.



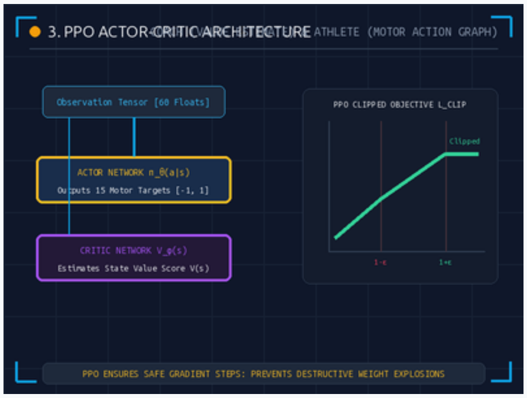
■ 2. Reward Engineering & Penalties

BALANCING PROGRESS VS ENERGY BURN

Mathematical Incentive Design: $R_t = w_{up} r_{up} + w_{fwd} r_{fwd} - w_\tau ||\tau||^2 - w_j ||\Delta a||^2$. The agent is penalized for high torque and action jerk.

The 12-Year-Old Analogy: If you only reward speed, the duck will throw itself down the stairs to cross the line. You must penalize falling and shaking.

- **Upright Bonus**: $r_{up} = +1.0$ if torso $z > 0.22\text{m}$ (immediate termination if $z < 0.14\text{m}$).
- **Velocity Tracking**: $r_{fwd} = \exp(-2.0 \cdot (v_x - v_{target})^2)$ matching target forward pace.
- **Torque Regularizer**: $-0.005 \cdot ||\tau||^2$ prevents motor overheating and gear wear.



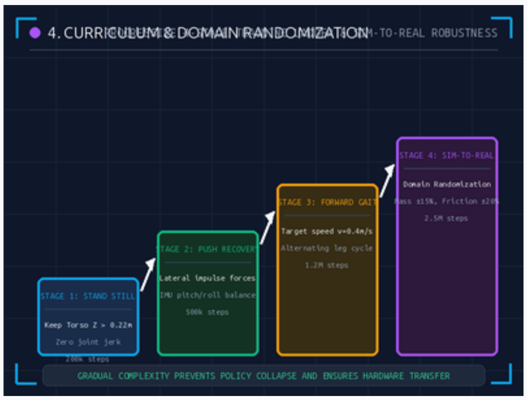
■ 3. PPO: Actor-Critic Architecture

COACH VS ATHLETE & CLIPPED OBJECTIVE

Proximal Policy Optimization: The Actor network $\pi_\theta(a|s)$ drives the joints, while the Critic network $V_\phi(s)$ evaluates performance. The clipped surrogate objective prevents destructive weight updates.

The 12-Year-Old Analogy: The Gymnast (Actor) performs a trick; the Coach (Critic) scores it. PPO stops the Gymnast from making crazy changes that erase past lessons.

- **Clipped Envelope**: $L^{CLIP}(\theta) = \hat{\mathbb{E}}_t [\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t)]$, with $\epsilon=0.2$.
- **Advantage Estimation**: GAE $(\lambda=0.95)$ computes temporal difference score $\hat{A}_t$ to guide policy gradient.
- **Pruned Edge Export**: Critic is discarded post-training; only lightweight Actor is exported to ONNX.



■ 4. Curriculum & Sim-to-Real Transfer

4-STAGE LADDER & DOMAIN RANDOMIZATION

Progressive Training: Locomotion is unlocked across 4 stages (Stand $\to$ Push Recovery $\to$ Gait $\to$ Domain Randomization) to avoid catastrophic policy collapse.

The 12-Year-Old Analogy: First teach a child to stand on two feet. Then teach balance against gentle shoves. Then take a step. Then run on wet grass.

- **Stage Progression**: Stage 1 (200k steps) $\to$ Stage 2 (500k steps) $\to$ Stage 3 (1.2M steps) $\to$ Stage 4 (2.5M steps).
- **Domain Randomization**: Perturbs link mass $(\pm 15\%)$, ground friction $(\pm 20\%)$, and latency $(1\dots 5\text{ms})$.
- **Hardware Safety Clamp**: torch.clamp(a, -1.0, 1.0) baked in silicon guarantees 0% gear stripping.

■ PHYSICAL AI COMPLETE MASTERCLASS CERTIFICATION

- **Hardware (Phase 1):** RK3566 onboard compute, 6-DOF IMU, 15 actuators, and battery constraints.
- **Simulation (Phase 2):** MuJoCo kinematic trees, collision geoms, autolimits, and 50Hz integration.
- **Control (Phase 3):** PPO reinforcement learning, clamped action graphs, and sim-to-real transfer.

■ PHASE 3 KNOWLEDGE CHECK

ASSESSMENT

Test your understanding of Reinforcement Learning, PPO, and Sim-to-Real transfer.

5 Questions

Q1: Temporal Observation Dimensions

MULTIPLE CHOICE

How many float values make up the Microduck's 4-frame observation tensor?

- **A.** 15 floats
- **B.** 60 floats (4 historical frames × 15 sensors)
- **C.** 1,000 floats

Q2: Torque Penalty in Reward Design

MULTIPLE CHOICE

Why must reward functions include a negative torque penalty term ($-||\tau||^2$)?

- **A.** To eliminate high-frequency motor shaking and conserve battery energy.
- **B.** Because torque does not exist in real robotics.
- **C.** To force the robot to walk backwards.

Q3: Actor-Critic Roles

MULTIPLE CHOICE

In Actor-Critic architectures like PPO, what is the specific role of the Critic?

- **A.** It directly sends PWM electrical commands to the motors.
- **B.** It acts as the Coach, predicting expected future discounted returns V(s).
- **C.** It regulates the cooling fans on the Rockchip CPU.

Q4: Silicon Safety Clamping

MULTIPLE CHOICE

Why is torch.clamp(a, -1.0, 1.0) baked directly into the ONNX graph?

- **A.** To mathematically guarantee motor targets never exceed physical limits, preventing gear stripping.
- **B.** To reduce the file size of the neural network.
- **C.** To activate the RGB camera lens.

Q5: Sim-to-Real Transfer

MULTIPLE CHOICE

What is Domain Randomization during curriculum training?

- **A.** Playing random background music during training.
- **B.** Randomly perturbing mass, friction, and latency in simulation so the policy transfers robustly to reality.
- **C.** Randomly shutting off power to the computer.

Answer Key: 1-B, 2-A, 3-B, 4-A, 5-B (1-B: 60 floats; 2-A: Energy/jerk suppression; 3-B: Value Coach; 4-A: Hardware protection; 5-B: Robust physics randomization)